STUDENT NAME: __Mr Answers ☺___

CHEN / QUINNELL

Prairiewood High School

2024 **TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION**

Mathematics Extension 1

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Write your name on each writing booklet

Total marks: 70

Section I – 10 marks (pages 3 - 7)

- Attempt Questions 1 - 10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 8 - 14)

- Attempt Questions 11 - 14
- Allow about 1 hour and 45 minutes for this section

MC	Q11	Q12	Q13	Q14	Total
/ 10	/ 15	/ 15	/ 15	/ 15	/ 70

DO NOT REMOVE ANY SECTION OF THIS EXAM FROM THE EXAMINATION ROOM

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SECTION I: Multiple Choice (10 marks)

Attempt Questions 1 – 10.

Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for this section.

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1. What is the remainder when the polynomial $P(x) = x^3 - 3ax$ is divided by $(x - a + 1)$?

(A) $a^3 + 1$

(B) $a^3 + 6a - 1$

(C) $a^3 - 6a^2 - 1$

(D) $a^3 - 6a^2 + 6a - 1$

Ans:

Remainder is $P(a-1) = (a-1)^3 - 3a(a-1) = a^3 - 6a^2 + 6a - 1$

2. A bag contains 6 blue balls, 7 green balls, 8 yellow balls, 9 orange balls and 10 red balls. Balls are drawn at random one at a time without replacement from the bag. What is the least number of balls that needs to be drawn from the bag in order to be certain that 8 balls of the same colour are amongst those selected?

(A) 21

(B) 22

(C) 35

(D) 36

Ans:

Max. number chosen without having 8 of same colour is 34 comprising 6 blue and 7 each of green, yellow, orange and red. Then 35th ball is one of yellow, orange or red giving 8 of one colour.

3. What is the Cartesian equation of the curve with the parametric equations below?

$$x = 3\sin\theta + 1$$
$$y = 3\cos\theta$$

(A) $x^2 + (y-1)^2 = 3$

(B) $(x-1)^2 + y^2 = 3$

(C) $x^2 + (y-1)^2 = 9$

(D) $(x-1)^2 + y^2 = 9$

Ans:

$x = 3\sin\theta + 1 \Rightarrow \sin\theta = \dfrac{x-1}{3}$

$y = 3\cos\theta \Rightarrow \cos\theta = \dfrac{y}{3}$

Now $\sin^2\theta + \cos^2\theta = 1$

$\left(\dfrac{x-1}{3}\right)^2 + \left(\dfrac{y}{3}\right)^2 = 1$

$\dfrac{(x-1)^2}{9} + \dfrac{y^2}{9} = 1$

$(x-1)^2 + y^2 = 9$

4. Let X be a random variable that follows a Binomial distribution with $E(X) = 7$ and $Var(X) = 6$. Which of the following could be the probability of success p?

(A) $\frac{1}{49}$

(B) $\frac{1}{7}$

(C) $\frac{36}{49}$

(D) $\frac{6}{7}$

Ans:

$$E(X) = np = 7$$
$$Var(X) = np(1-p) = 6$$

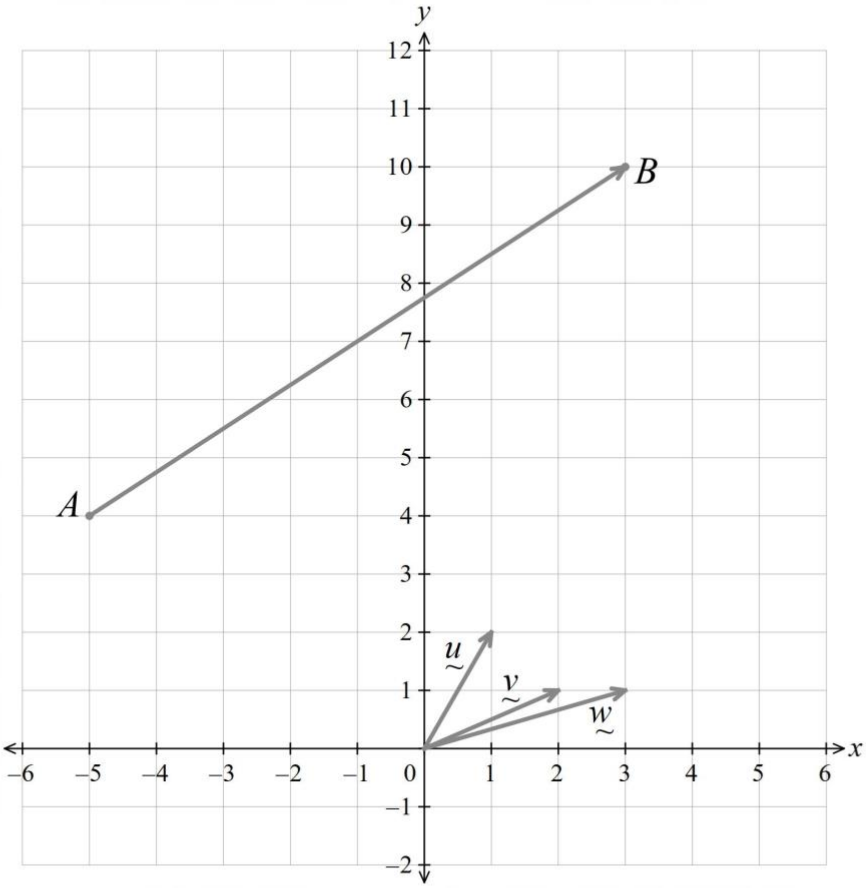
Solving simultaneously, we find

$$7(1-p) = 6$$
$$p = \frac{1}{7}$$

5. On the diagram below, the vector $\overrightarrow{AB}$ is shown joining $A(-5,4)$ and $B(3,10)$.

There are three other vectors shown on the diagram:

$$\underset{\sim}{u} = \underset{\sim}{i} + 2\underset{\sim}{j} \qquad \underset{\sim}{v} = 2\underset{\sim}{i} + \underset{\sim}{j} \qquad \underset{\sim}{w} = 3\underset{\sim}{i} + \underset{\sim}{j}$$



Which of the following vectors is parallel to $\overrightarrow{AB}$?

(A) $\underset{\sim}{u} - \underset{\sim}{v}$

(B) $\underset{\sim}{u} + \underset{\sim}{v}$

(C) $\underset{\sim}{w} + \underset{\sim}{v}$

(D) $\underset{\sim}{w} + \underset{\sim}{u}$

Ans:

Simplest solution is to sketch the 4 resultant vectors and it is clear that $\underset{\sim}{w} + \underset{\sim}{u}$ is parallel to $\overrightarrow{\text{AB}}$.

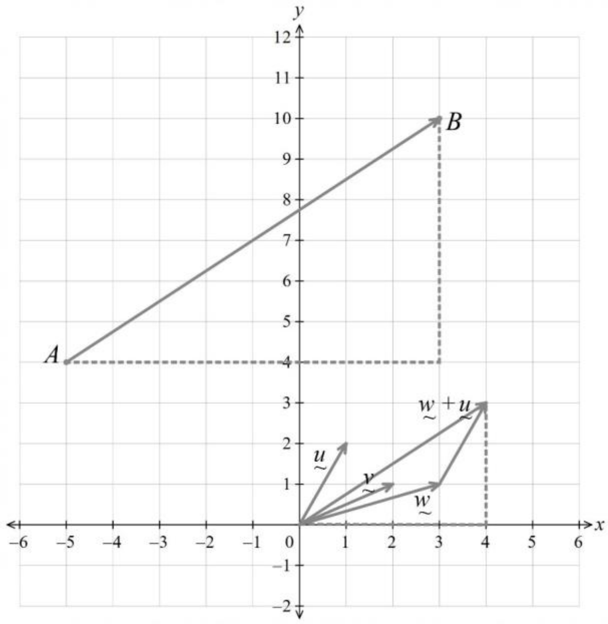
$\underset{\sim}{w} + \underset{\sim}{u}$ is inclined at

$\theta = \tan^{-1}\left(\frac{3}{4}\right)$

$\overrightarrow{\text{AB}}$ is inclined at

$\alpha = \tan^{-1}\left(\frac{6}{8}\right) = = \tan^{-1}\left(\frac{3}{4}\right) = \theta$

so the vectors are parallel



Alternative Solution:

$\overrightarrow{AB} = \begin{pmatrix} 8 \\ 6 \end{pmatrix}$ Gradient $= \frac{6}{8} = \frac{3}{4}$

$\underset{\sim}{u} - \underset{\sim}{v} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

$\underset{\sim}{u} + \underset{\sim}{v} = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$

$\underset{\sim}{w} + \underset{\sim}{v} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$

$\underset{\sim}{w} + \underset{\sim}{u} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$ Gradient $= \frac{3}{4}$

Only $\underset{\sim}{w} + \underset{\sim}{u}$ has the same gradient as $\overrightarrow{AB}$ so the vectors are parallel

Further alternative:

Can find scalar product $\overrightarrow{AB}.(\underset{\sim}{w} + \underset{\sim}{u}) = 8 \times 4 + 6 \times 3 = 50$

So $|\overrightarrow{AB}|.|\underset{\sim}{w} + \underset{\sim}{u}|.\cos\theta = 50$

$10 \times 5\cos\theta = 50$

$\cos\theta = 1$

$\therefore\ \theta = 0$

Angle between the vectors is zero so they are parallel

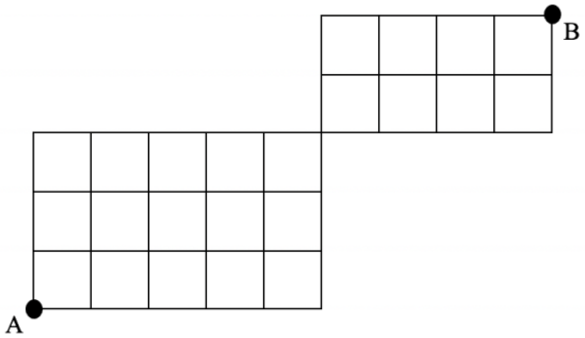
6. Moving only right or up, how many different paths exist to get from Point A to Point B ?

(A) 45

(B) 840

(C) 480480

(D) 16!



Ans:

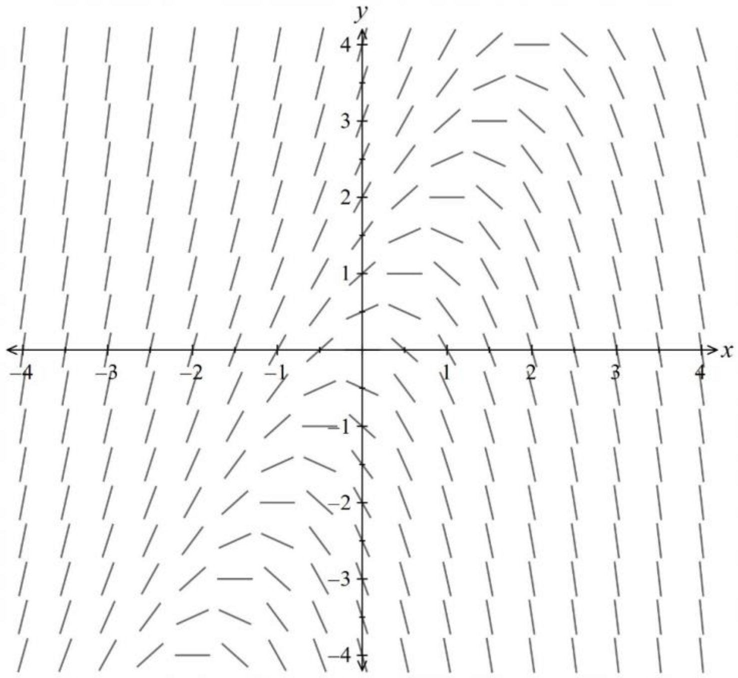
The diagram can be split into two separate rectangles as they only connect at a corner.

In the first rectangle, there are a total of 5 rights and 3 ups required to get to the upper right corner. This can be done in $\frac{8!}{5!3!}$ ways.

In the second rectangle, there are a total of 4 rights and 2 ups required to get to the upper right corner. This can be done in $\frac{6!}{4!2!}$ ways.

The total number of paths will be $\frac{8!}{5!3!} \times \frac{6!}{4!2!} = 840$.

7. The direction field for a differential equation is shown below.



Considering the point $(1, 1)$ determine which of the differential equations below could give this direction field?

(A) $y' = (y - 2x)$

(B) $y' = (y + 2x)$

(C) $y' = (y + x)$

(D) $y' = (x - y)$

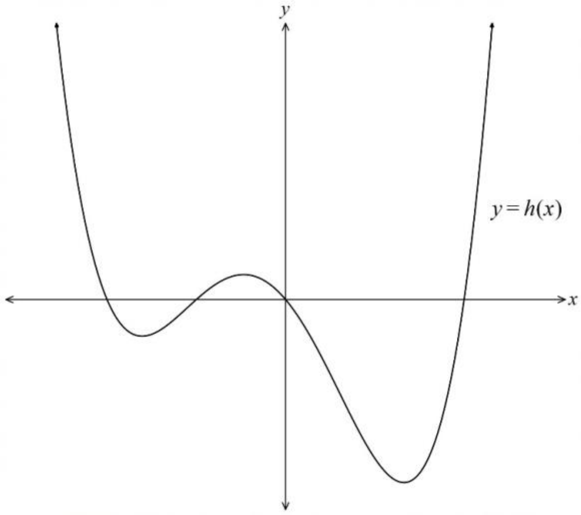
Ans:

Take the point (1, 1) (which has a negative gradient in the direction field) and test with the four equations.

$A.\, y' = (y - 2x) \quad y' = (1 - 2(1)) = -1$ (− gradient)
$B.\, y' = (y + 2x) \quad y' = (1 + 2(1)) = 3$ (+gradient)
$C.\, y' = (y + x) \quad y' = (1 + 1) = 2$ (+ gradient)
$D.\, y' = (x - y) \quad y' = 1 - 1 = 0$ (Horizontal gradient)

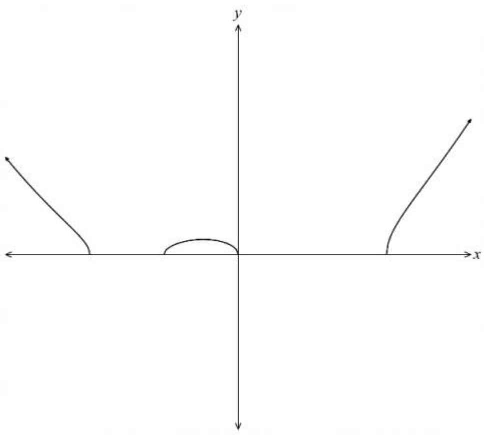
Only A has a negative gradient at the point (1, 1).

8. The graph of $y = h(x)$ is shown below.

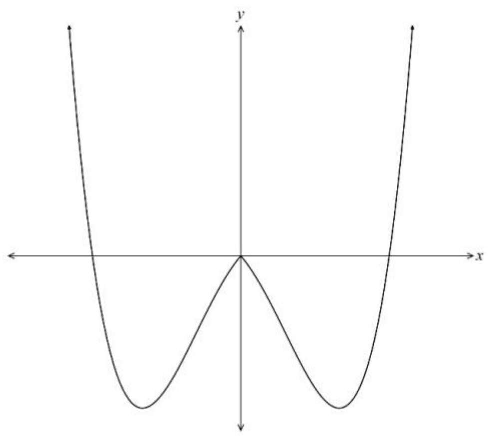


Which graph shows $y = |h(x)|$?

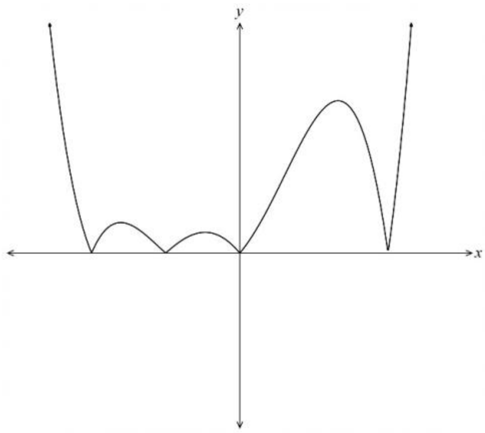
(A)



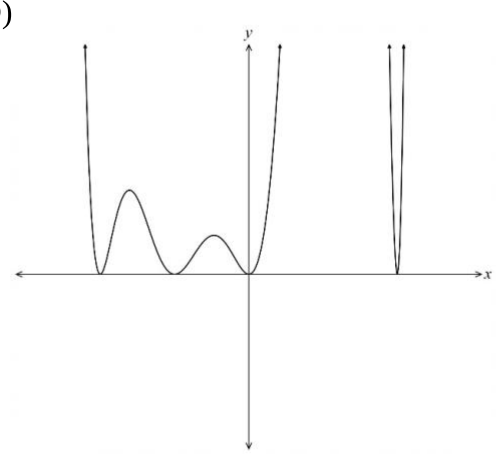
(B)



(C)



(D)



Ans:

The sections where the curve dips below the x-axis are reflected in the x-axis (the negative y-values become positive when the absolute value is taken), while those above the x-axis remain unchanged.

9. If $y = \tan^{-1}\left(\frac{1}{x}\right)$, which of the following is an expression for $\frac{dy}{dx}$?

(A) $\frac{dy}{dx} = -\sin^2 y$

(B) $\frac{dy}{dx} = -\cos^2 y$

(C) $\frac{dy}{dx} = -\tan^2 y$

(D) $\frac{dy}{dx} = -\cot^2 y$

Ans:

$$y = \tan^{-1}\left(\frac{1}{x}\right) \qquad \frac{dy}{dx} = \frac{1}{1+\left(\frac{1}{x}\right)^2}\left(-\frac{1}{x^2}\right) = -\frac{\tan^2 y}{1+\tan^2 y}$$

$$\therefore \quad \frac{dy}{dx} = -\frac{\sin^2 y}{\cos^2 y \sec^2 y} = -\sin^2 y$$

10. What is the range of the function $y = 2\sin x + \cos x - 1$?

(A) $[-1-\sqrt{5}, -1+\sqrt{5}]$

(B) $[1-\sqrt{5}, 1+\sqrt{5}]$

(C) $[-\sqrt{5}, \sqrt{5}]$

(D) $[-3,1]$

Ans:

Using the auxiliary angle method for the two trig functions:

$$y = 2\sin x + \cos x - 1$$
$$y = \sqrt{5}\sin(x+\alpha) - 1 \text{ for some } \alpha$$

The max and min values are $-1-\sqrt{5}$ and $-1+\sqrt{5}$ respectively.

End of Section I

STUDENT NUMBER/NAME: ………………………….

SECTION II (60 marks)

Attempt Questions 11 – 14.

Allow about 1 hours and 45 minutes for this section.

Answer each question in the appropriate writing booklet. Extra booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) **Start a new writing booklet.**

(a) Find the number of ways that the letters of the word TRIANGLE can be arranged in a line such that the vowels are all next to each other but the consonants are not all next to each other. **2**

Arrange I, A, E (in any order)
T, R, N, G, L in a straight line so that the group of vowels is neither first nor last.
This can be done in $4 \times 3! \times 5! = 2880$ ways.

(b) Prove that $\frac{\sin\theta}{\cos\theta+\sin\theta} + \frac{\sin\theta}{\cos\theta-\sin\theta} = \tan(2\theta)$. **2**

$$LHS = \frac{\sin\theta}{\cos\theta + \sin\theta} + \frac{\sin\theta}{\cos\theta - \sin\theta}$$

$$= \frac{\sin\theta\,(\cos\theta - \sin\theta) + \sin\theta\,(\cos\theta + \sin\theta)}{(\cos\theta + \sin\theta)(\cos\theta - \sin\theta)}$$

$$= \frac{2\sin\theta\cos\theta}{\cos^2\theta - \sin^2\theta}$$

$$= \frac{\sin 2\theta}{\cos 2\theta}$$

$$= \tan 2\theta.$$

(c) Find the exact value of the integral below: **2**

$$\int_0^1 \frac{2x^2+1}{25+(2x^3+3x)^2}\,dx$$

$$\begin{aligned}
\int_0^1 \frac{2x^2+1}{25+(2x^3+3x)^2}\,dx &= \frac{1}{3}\int_0^1 \frac{6x^2+3}{5^2+(2x^3+3x)^2}\,dx \\
&= \frac{1}{3}\int_0^1 \frac{\frac{d}{dx}(2x^3+3x)}{5^2+(2x^3+3x)^2}\,dx \\
&= \frac{1}{3}\left[\frac{1}{5}\tan^{-1}\left(\frac{2x^3+3x}{5}\right)\right]_0^1 \\
&= \frac{1}{15}\left[\tan^{-1}(1)-\tan^{-1}(0)\right] \\
&= \frac{1}{15}\left[\frac{\pi}{4}-0\right] \\
&= \frac{\pi}{60}
\end{aligned}$$

(d) Use the $t=\tan\frac{x}{2}$ results to show that $\sin x-\tan\frac{x}{2}=\tan\frac{x}{2}\cos x$ **2**

$$\begin{aligned}
t=\tan\frac{x}{2} \Rightarrow \sin x-\tan\frac{x}{2} &= \frac{2t}{1+t^2}-t \\
&= t\left(\frac{2-(1+t^2)}{1+t^2}\right) \\
&= t\left(\frac{1-t^2}{1+t^2}\right) \\
&= \tan\frac{x}{2}\cos x
\end{aligned}$$

(e) Given α, β and γ are the roots of the equation $4x^3 + 5x^2 + 4x - 1 = 0$, find the value of: **2**

$$\frac{1}{\alpha^2\beta^2} + \frac{1}{\beta^2\gamma^2} + \frac{1}{\alpha^2\gamma^2}$$

$$\begin{aligned}\frac{1}{\alpha^2\beta^2} + \frac{1}{\beta^2\gamma^2} + \frac{1}{\alpha^2\gamma^2} &= \frac{\alpha^2 + \beta^2 + \gamma^2}{(\alpha\beta\gamma)^2} \\ &= \frac{(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)}{(\alpha\beta\gamma)^2} \\ &= \frac{\left(\frac{-5}{4}\right)^2 - 2\left(\frac{4}{4}\right)}{\left(\frac{1}{4}\right)^2} \\ &= -7\end{aligned}$$

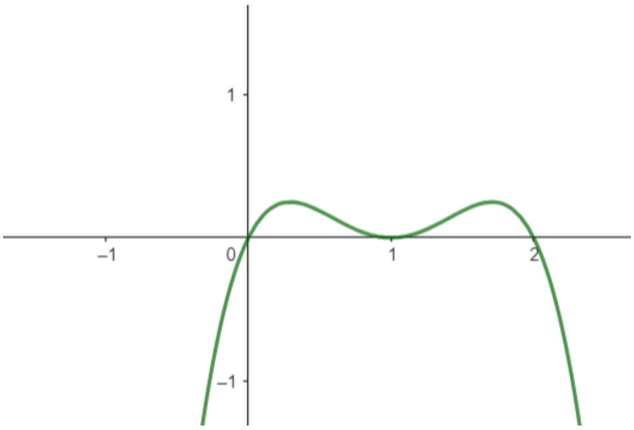
(f) Solve $\frac{1}{2x-x^2} \geq 1$. **3**

$$\frac{1}{2x - x^2} \geq 1$$
$$\frac{1}{x(2 - x)} - 1 \geq 0$$
$$\frac{1 - 2x + x^2}{x(2 - x)} \geq 0$$

multiplying both sides by the square of the denominator

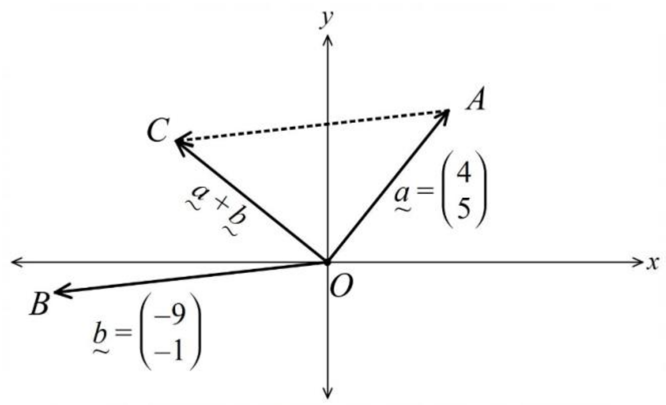
$x(1 - x)^2(2 - x) \geq 0$

Which is a negative quartic with roots $x = 0, 1, 2$.



The solution for the inequality will be $0 < x < 2$, since $x \neq 0,2$.

(g) The diagram shows the vectors $\underset{\sim}{a} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$, $\underset{\sim}{b} = \begin{pmatrix} -9 \\ -1 \end{pmatrix}$ and the resultant vector $\underset{\sim}{a} + \underset{\sim}{b}$. **2**



Show that ΔOAC, formed by the vectors, is an isosceles triangle and find the size of $\angle OAC$.

$\underset{\sim}{a} + \underset{\sim}{b} = \begin{pmatrix} 4-9 \\ 5-1 \end{pmatrix} = \begin{pmatrix} -5 \\ 4 \end{pmatrix}$

$|\underset{\sim}{a}| = \sqrt{4^2 + 5^2} = \sqrt{41}$

$|\underset{\sim}{a} + \underset{\sim}{b}| = \sqrt{(-5)^2 + 4^2} = \sqrt{41}$

$\therefore\ |\underset{\sim}{a}| = |\underset{\sim}{a} + \underset{\sim}{b}|$

$\therefore\ \angle OAC = \angle OCA$ (Base angles of isosceles triangle)

$(\underset{\sim}{a}) . (\underset{\sim}{a} + \underset{\sim}{b}) = |\underset{\sim}{a}| \times |\underset{\sim}{a} + \underset{\sim}{b}| \times \cos(\angle AOC)$

$4 \times (-5) + 5 \times 4 = \sqrt{41} \times \sqrt{41} \times \cos(\angle AOC)$

$-20 + 20 = 41\cos(\angle AOC)$

$\cos(\angle AOC) = 0$

$\angle AOC = 90°$

$\therefore\ \angle OAC = \angle OCA = \dfrac{90}{2} = 45°$

Alternate Solution

$|\underset{\sim}{b}| = \sqrt{(-9)^2 + (-1)^2} = \sqrt{82}$

$(\underset{\sim}{a}) . (\underset{\sim}{b}) = |\underset{\sim}{a}| \times |\underset{\sim}{b}| \times \cos(\angle AOB)$

$4 \times (-9) + 5 \times (-1) = \sqrt{82} \times \sqrt{41} \times \cos(\angle AOB)$

$-36 - 5 = 41\sqrt{2}\cos(\angle AOB)$

$-41 = 41\sqrt{2}\cos(\angle AOB)$

$\cos(\angle AOB) = -\dfrac{1}{\sqrt{2}}$

$\angle AOB = \arccos\left(-\dfrac{1}{\sqrt{2}}\right) = 135°$

$\angle AOB + \angle OAC = 180°$ (cointerior angles)

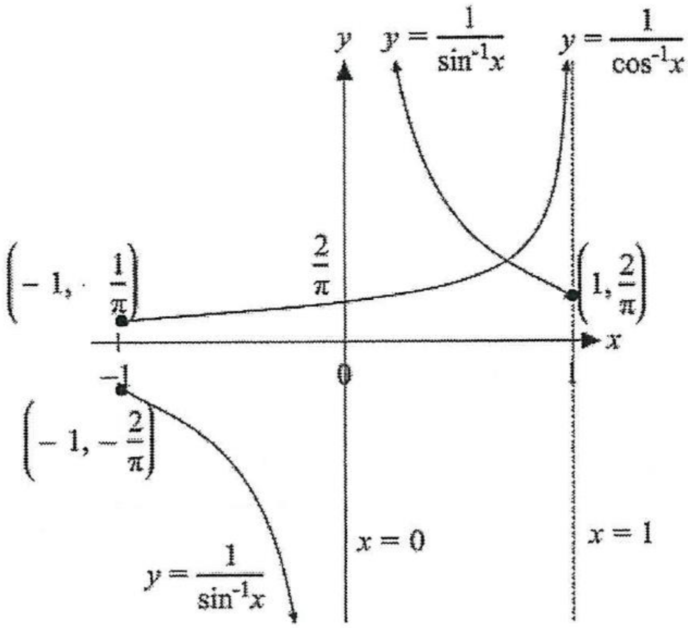
$\angle OAC = 180° - 135° = 45° = \angle OCA$

End of Question 11

Question 12 (15 marks) **Start a new writing booklet.**

(a) On the same diagram, sketch the curves $y = \frac{1}{\sin^{-1} x}$ and $y = \frac{1}{\cos^{-1} x}$, **3**

clearly showing any intercepts, endpoints and the equations of any asymptotes.



(b) Given that $P(x) = x^4 - 9x^3 + 25x^2 - 27x + 10$, has a double root, factorise $P(x)$ into linear factors. **2**

The roots must be factors of 10
i.e. $x = \pm 1, \pm 2, \pm 5$
$x = +1$ or -1 must be the double root, as 1^2 is a factor of 10, but 2^2 and 5^2 are not.

$P(1) = (1)^4 - 9(1)^3 + 25(1)^2 - 27(1) + 10 = 0$
So $(x-1)^2 = x^2 - 2x + 1$ is a factor

Could use division to get $P(x) = (x^2 - 2x + 1)(x^2 - 7x + 10)$
Then factorise to obtain $P(x) = (x-1)^2(x-2)(x-5)$
Or test the factors ± 2 and ± 5
$P(2) = (2)^4 - 9(2)^3 + 25(2)^2 - 27(2) + 10 = 0$
$P(5) = (5)^4 - 9(5)^3 + 25(5)^2 - 27(5) + 10 = 0$

So $(x-2)$ and $(x-5)$ are factors
So 14actorization is:
$P(x) = (x-1)^2(x-2)(x-5)$

(c) A thin sheet of ice is in the form of a circle. If the ice is melting in such a way that the area of the sheet is decreasing at a rate of $0.5\ \text{m}^2/\text{s}$. At what rate is the radius decreasing when the area of the sheet is $12\ \text{m}^2$? **3**

For a circle we have $A = \pi r^2$ which gives $\frac{dA}{dr} = 2\pi r$. If $A = 12, r = \sqrt{\frac{12}{\pi}}$

Since $\frac{dA}{dt} = -\frac{1}{2}$ we can find

$$\begin{aligned}\frac{dr}{dt} &= \frac{dr}{dA} \times \frac{dA}{dt} \\ &= \frac{1}{2\pi\sqrt{\frac{12}{\pi}}} \times -\frac{1}{2} \\ &= -0.0407\ (4\ \text{d.p.})\end{aligned}$$

(d) Express $\sqrt{6}\sin x + \sqrt{2}\cos x$ in the form $R\sin(x+\alpha)$ and hence or otherwise, solve: **3**

$$\sqrt{6}\sin x + \sqrt{2}\cos x = 2, \qquad [0, 2\pi]$$

Let $\sqrt{6}\sin x + \sqrt{2}\cos x = R\sin(x+\alpha)$

Then $\tan\alpha = \frac{\sqrt{2}}{\sqrt{6}} = \frac{1}{\sqrt{3}}$ and $R^2 = \left(\sqrt{6}\right)^2 + \left(\sqrt{2}\right)^2 = 8$

$\alpha = \arctan\left(\frac{1}{\sqrt{3}}\right)$ and $R^2 = 2 + 6 = 8$

$\alpha = \frac{\pi}{6}$ and $R = \sqrt{8}$

$\sqrt{6}\sin x + \sqrt{2}\cos x = \sqrt{8}\sin\left(x + \frac{\pi}{6}\right)$

$$\sqrt{6}\sin x + \sqrt{2}\cos x = 2$$

$$\therefore \quad \sqrt{8}\sin\left(x + \frac{\pi}{6}\right) = 2$$

$$\sin\left(x + \frac{\pi}{6}\right) = \frac{2}{\sqrt{8}} = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$x + \frac{\pi}{6} = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{9\pi}{4}, \frac{11\pi}{4} \ \ldots..$$

$$x = \frac{\pi}{4} - \frac{\pi}{6}, \frac{3\pi}{4} - \frac{\pi}{6}, \frac{9\pi}{4} - \frac{\pi}{6}, \frac{11\pi}{4} - \frac{\pi}{6} \ \ldots.$$

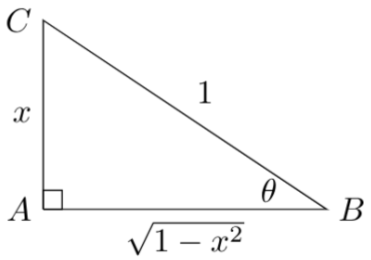
$$x = \frac{\pi}{12}, \frac{7\pi}{12}, \frac{25\pi}{12}, \frac{31\pi}{12} \ \ldots.$$

For required domain of $0 \le x \le 2\pi$

$$x = \frac{\pi}{12}, \frac{7\pi}{12}$$

(e) Show that $\sin^{-1} x + \sin^{-1}\sqrt{1-x^2} = \frac{\pi}{2}$. **2**

Let $\theta = \sin^{-1} x$ which allows us to draw:



From the diagram, we observe $\sin^{-1}\sqrt{1-x^2} = \cos^{-1} x$, hence

$$\sin^{-1} x + \sin^{-1}\sqrt{1-x^2} = \sin^{-1} x + \cos^{-1} x$$

$$= \frac{\pi}{2}$$

(f) Using the substitution $u = -\frac{1}{x}$, evaluate **2**

$$\int_1^2 \frac{1}{x^2} e^{-\frac{1}{x}}\, dx.$$

Let $u = -\frac{1}{x}$ hence $\frac{du}{dx} = \frac{1}{x^2}$. When $x = 1$ and 2, $u = -1$ and $-\frac{1}{2}$ respectively.

$$\int_1^2 \frac{1}{x^2} e^{-\frac{1}{x}}\, dx = \int_{-1}^{-\frac{1}{2}} e^u\, du$$

$$= [e^u]_{-1}^{-\frac{1}{2}}$$

$$= e^{-\frac{1}{2}} - e^{-1}$$

End of Question 12

Question 13 (15 marks) **Start a new writing booklet.**

(a) A box initially contains 100 grams of bacteria and 10 000 grams of a radioactive material with a half-life of 3 hours. An hour later, the mass of bacteria has grown to 150 grams. Assuming exponential growth and decay, what time (t) will the mass of bacteria and radioactive material be the same? Leave your answer correct to 2 decimal places. **3**

Since we are modelling exponential growth and decay, we can use and decay, for mass of bacteria, B, and mass of radioactive material, M, and time in hours, t to write the equations:

$B = 100e^{k_1 t}$ and $M = 10000e^{k_2 t}$.

The half-life of M is 3 hours so we can solve for k_2:

$$\frac{1}{2} = e^{3k_2}$$

$$k_2 = \frac{1}{3}\ln\frac{1}{2}$$

$$= -\frac{1}{3}\ln 2$$

The Bacteria growth after 1 hour lets us solve for k_1:

$$150 = 100e^{k_1}$$

$$k_1 = \ln\frac{3}{2}$$

Solving for t when $B = M$ gives:

$$100e^{\ln\frac{3}{2}t} = 10000e^{-\frac{1}{3}\ln(2)t}$$

$$100 = e^{\left(\ln\frac{3}{2}+\frac{1}{3}\ln 2\right)t}$$

$$t = \frac{\ln 100}{\ln\frac{3}{2} + \frac{1}{3}\ln 2}$$

$$= 7.23 \text{ hours}$$

(b) Use the method of separation of variables to solve the differential equation: $\frac{dy}{dx} = \frac{3y}{x}$, **2**

given that $\frac{dy}{dx} = 6$ when $x = 2$.

Method 1

$$\frac{dy}{dx} = \frac{3y}{x}$$
$$\frac{dy}{3y} = \frac{dx}{x}$$
$$\frac{1}{3}\int \frac{dy}{y} = \int \frac{dx}{x}$$
$$\frac{1}{3}\ln y = \ln x + c_1$$
$$\ln y = 3\ln x + c_2$$
$$\ln y = \ln x^3 + \ln c_3$$
$$\ln y = \ln (cx^3)$$
$$y = cx^3$$

When $x = 2, \frac{dy}{dx} = 6$

$$6 = \frac{3y}{2}$$
$$3y = 12$$
$$y = 4$$
$$y = cx^3$$
$$4 = c \times 2^3$$
$$c = \frac{4}{8} = \frac{1}{2}$$

Solution

$$y = \frac{x^3}{2}$$

Method 2

$$\frac{dy}{dx} = \frac{3y}{x}$$
$$\frac{dy}{3y} = \frac{dx}{x}$$
$$\frac{1}{3}\int \frac{dy}{y} = \int \frac{dx}{x}$$
$$\frac{1}{3}\ln y = \ln x + c_1$$

at (2,4) $\quad \frac{1}{3}\ln 4 = \ln 2 + c_1$

$$c_1 = \frac{1}{3}\ln 4 - \ln 2$$

$$\frac{1}{3}\ln y = \ln x + \frac{1}{3}\ln 4 - \ln 2$$
$$\ln y = 3\ln x + \ln 4 - 3\ln 2$$
$$= \ln x^3 + \ln 4 - \ln 8$$
$$= \ln\left(\frac{4x^3}{8}\right)$$
$$\ln y = \ln\left(\frac{x^3}{2}\right)$$
$$y = \frac{x^3}{2}$$

(c) Records show that 40% of residents of a town went to school within the town. A sample of 150 residents of the town is to be taken to determine the population who went to school within the town.

(i) Find, in simplest fraction form, the mean and standard deviation of the distribution of such sample proportions. **2**

Distribution of sample proportion $\widehat{p}$ has $\mu = \frac{2}{5}$ and $\sigma = \sqrt{\dfrac{\frac{2}{5} \times \frac{3}{5}}{150}} = \frac{1}{25}$

(ii) Use the following extract of the table of values of $P(Z \leq z)$, where Z has a standard normal distribution, to estimate correct to 2 decimal places the probability that a sample of 150 residents of the town contains at least 57 and at most 69 who went to school in town. **3**

STANDATD NORMAL DISTRIBUTION TABLE

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767

Ignoring the continuity correction, $P\left(\frac{57}{150} \leq \widehat{p} \leq \frac{69}{150}\right) \approx P\left(\dfrac{\frac{19}{50} - \frac{2}{5}}{\frac{1}{25}} \leq Z \leq \dfrac{\frac{23}{50} - \frac{2}{5}}{\frac{1}{25}}\right) = P\left(-\frac{1}{2} \leq Z \leq \frac{3}{2}\right)$

If $\Phi(z) = P(Z \leq z)$, $P\left(-\frac{1}{2} \leq Z \leq \frac{3}{2}\right) = \Phi(1 \cdot 5) - (1 - \Phi(0 \cdot 5)) = \Phi(1 \cdot 5) + \Phi(0 \cdot 5) - 1$

$\therefore P\left(\frac{57}{150} \leq \widehat{p} \leq \frac{69}{150}\right) \approx 0 \cdot 9332 + 0 \cdot 6915 - 1 \approx 0 \cdot 62$

(d) Use mathematical induction to prove that $9^n - 4^n$ is divisible by 5 for all integers $n \geq 1$. **3**

Show that it is true for $n = 1$
$9^1 - 4^1 = 9 - 4 = 5 = 5 \times 1$
$\therefore$ true for $n = 1$

Assume that it is true for $n = k$
i.e. that $9^k - 4^k = 5p$ where p is an integer

Show that it is true for $n = k + 1$
i.e. that $9^{k+1} - 4^{k+1} = 5q$ where q is an integer

$$\begin{aligned}
\text{LHS} &= 9^{k+1} - 4^{k+1} \\
&= 9 \times 9^k - 4 \times 4^k \\
&= 9 \times 9^k - 4 \times 4^k - 5 \times 4^k + 5 \times 4^k \\
&= 9 \times 9^k - 9 \times 4^k + 5 \times 4^k \\
&= 9(9^k - 4^k) + 5 \times 4^k \\
&= 9 \times 5p + 5 \times 4^k \\
&= 5(9p + 4^k) \\
&= 5q \text{ since } 9p + 4^k \text{ is an integer, as } p \text{ and } k \text{ are integers}
\end{aligned}$$

$\therefore$ If true for for $n = k$, it is also true for $n = k + 1$. Since true for $\mathrm{n} = 1$, then by PMI, it is true for all integers $n \geq 1$.

(e) Given that ${}^{n}\mathrm{C}_4 = p$ and ${}^{n}\mathrm{C}_{5} = q$, use the Pascals Triangle relationships between coefficients to find an expression, in terms of p and q, for ${}^{n+1}\mathrm{C}_{n-5} + {}^{n}\mathrm{C}_{n-4}$. **2**

${}^{n}\mathbf{C}_4 = p$ and ${}^{n}\mathbf{C}_5 = q$
${}^{n}\mathbf{C}_{n-4} = p$ (Symmetry of Pascals Triangle)
${}^{n}\mathbf{C}_4 + {}^{n}\mathbf{C}_5 = {}^{n+1}\mathbf{C}_5 = p + q$ (Sums of terms below in P Triangle)
${}^{n+1}\mathbf{C}_5 = {}^{n+1}\mathbf{C}_{n-5} = p + q$ (Symmetry of Pascals Triangle)

$$\begin{aligned}
{}^{n+1}\mathbf{C}_{n-5} + {}^{n}\mathbf{C}_{n-4} &= p + q + p \\
&= 2p + q
\end{aligned}$$

End of Question 13

Question 14 (15 marks)   **Start a new writing booklet.**

(a) In the expansion of the expression $(1+x)(a-bx)^{12}$, the coefficient of x^8 is zero. Find $\frac{a}{b}$ in simplest form provided neither a or b are zero. **2**

In the expansion of $(1+x)(a-bx)^{12}$, the coefficient of the x^8 term is given by:

$$\binom{12}{8}a^4(-b)^8 + \binom{12}{7}a^5(-b)^7 = 0$$

$$a^4b^7\left(\frac{12!}{8!\,4!}b - \frac{12!}{7!\,5!}a\right) = 0$$

$$\frac{12!}{8!\,4!}b = \frac{12!}{7!\,5!}a$$

$$\frac{a}{b} = \frac{5}{8}.$$

(b) Use Mathematical Induction to show that for all integers $n \geq 1$ **3**

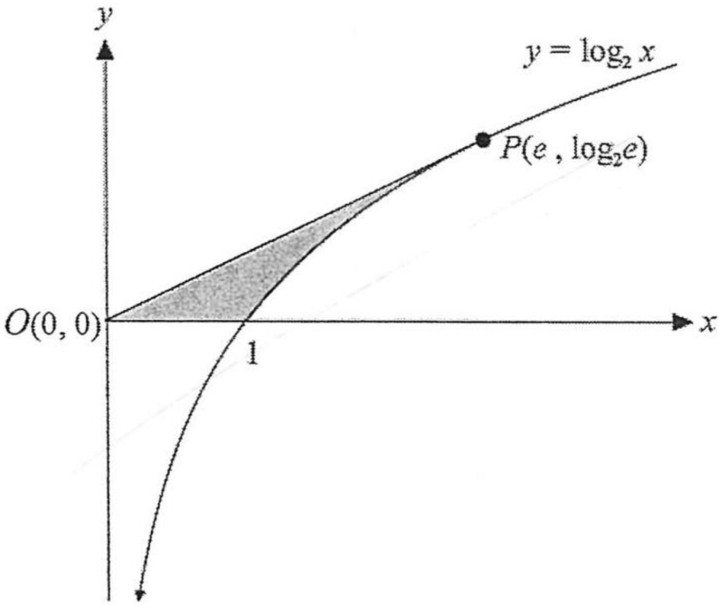
$$\left(1 \times 1! \times \frac{0}{1!}\right) + \left(2 \times 2! \times \frac{1}{2!}\right) + \left(3 \times 3! \times \frac{2}{3!}\right) + \cdots + \left(n \times n! \times \frac{n-1}{n!}\right) = (n+1)! - \frac{1}{n!}$$

Error in question.

All students will receive 3 marks for this question.

(c) The diagram shows the point $P(e, \log_2 e)$ on the curve $y = \log_2 x$. **3**

OP is a tangent to the curve at the point P.



The shaded region bounded by the curve $y = \log_2 x$, the line OP and the x axis is rotated through one revolution about the y axis. Find, in simplest exact form, the volume of the solid formed.

Volume is given by $V = \pi \int_0^{\log_2 e} 2^{2y}\,dy - \frac{1}{3}\pi e^2 \log_2 e$

$$= \frac{\pi}{2\ln 2}\Big[\ 2^{2y}\ \Big]_0^{\log_2 e} - \frac{\pi e^2}{3}\log_2 e$$

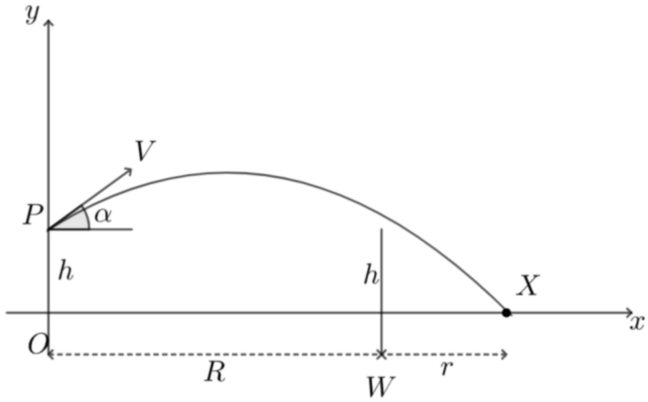
$$= \frac{\pi \ln e}{2\ln 2}\left(2^{2\log_2 e} - 1\right) - \frac{\pi e^2}{3}\log_2 e$$

$$= \pi \log_2 e\left\{\tfrac{1}{2}\left(e^2 - 1\right) - \tfrac{1}{3}e^2\right\}$$

$$= \pi \log_2 e\left(\frac{1}{6}e^2 - \frac{1}{2}\right)$$

Hence volume is $\frac{1}{6}\pi\left(e^2 - 3\right)\log_2 e$ cubic units.

(d) A projectile is launched from point P at a height of h metres above the ground with velocity V m/s at an angle α to the horizontal towards the point X. There is a wall with height h metres at point W, which is R metres from the point O.



The position vector of the projectile, t seconds after it is projected and with acceleration due to gravity, g, is given by

$$\mathrm{r}(t) = \begin{pmatrix} Vt\cos\alpha \\ -\frac{1}{2}gt^2 + Vt\sin\alpha + h \end{pmatrix}$$

DO NOT prove this.

(i) The projectile just clears the wall. Show that

$$V^2 \geq \frac{gR}{2\sin\alpha\cos\alpha}.$$

3

Given $\mathbf{r}(t) = \begin{pmatrix} Vt\cos\alpha \\ -\frac{1}{2}gt^2 + Vt\sin\alpha + h \end{pmatrix}$, we can find the Cartesian equation by eliminating t.

Using $x = Vt\cos\alpha$ we see $t = \frac{x}{V\cos\alpha}$ which can be substituted into $y = -\frac{1}{2}gt^2 + Vt\sin\alpha + h$ to give:

$$\begin{aligned} y &= -\frac{1}{2}g\left(\frac{x}{V\cos\alpha}\right)^2 + V\left(\frac{x}{V\cos\alpha}\right)\sin\alpha + h \\ &= h + x\tan\alpha - x^2\frac{g}{2V^2\cos^2\alpha}. \end{aligned}$$

Since the projectile clears the wall, we know that when $x = R$, $y \geq h$. Thus we can solve,

$$\begin{aligned} h + R\tan\alpha - R^2\frac{g}{2V^2\cos^2\alpha} &\geq h \\ R\tan\alpha &\geq R^2\frac{g}{2V^2\cos^2\alpha} \\ V^2 &\geq \frac{gR}{2\sin\alpha\cos\alpha} \end{aligned}$$

(ii) After clearing the wall, the projectile hits the ground at the point X. Show that

$$\tan\alpha \geq \frac{Rh}{(R+r)r}$$

Since the projectile hits the ground at point X, we have $x = R + r$ when $y = 0$, thus:

$$0 = h + (R+r)\tan\alpha - (R+r)^2\frac{g}{2V^2\cos^2\alpha}$$

Now we also know that $V^2 \geq \frac{gR}{2\sin\alpha\cos\alpha}$ since it clears the wall so we can simplify the above expression to:

$$0 \geq h + (R+r)\tan\alpha - (R+r)^2\frac{g}{2\left(\frac{gR}{2\sin\alpha\cos\alpha}\right)\cos^2\alpha}$$

$$= h + (R+r)\tan\alpha - \frac{(R+r)^2}{\left(\frac{R}{\sin\alpha}\right)\cos\alpha}$$

$$= h + (R+r)\tan\alpha - \frac{(R+r)^2}{R}\tan\alpha$$

$$= h + (R+r)\tan\alpha\left(1 - \frac{(R+r)}{R}\right)$$

$$-(R+r)\tan\alpha\left(1 - \frac{(R+r)}{R}\right) \geq h$$

$$(R+r)\tan\alpha\left(\frac{(R+r)}{R} - 1\right) \geq h$$

Note: $\frac{(R+r)}{R} - 1$ is positive as $R, r > 0$

$$\tan\alpha \geq \frac{h}{(R+r)\left(\frac{(R+r)}{R} - 1\right)}$$

$$\tan\alpha \geq \frac{h}{\frac{(R+r)(R+r-R)}{R}}$$

$$\tan\alpha \geq \frac{Rh}{r(R+r)} \text{ as required}$$

(iii) Suppose the projectile clears the wall, and $V \leq 50$, $g = 10, R = 80$, and $h = 1$. **2**
What is the closest point to the wall where the projectile can land?

To find the closest point the projectile can land after clearing the wall, we have the conditions: $V \leq 50$ which implies $V^2 \leq 2500$, $g = 10, R = 80$.

From Part (i) we now see:

$$2500 \geq \frac{(10)(80)}{2\sin\alpha\cos\alpha}$$
$$2\sin\alpha\cos\alpha \geq \frac{800}{2500}$$
$$\sin 2\alpha \geq \frac{8}{25}$$

Which has solution: $9.33° \leq \alpha \leq 80.67°$.

To minimise the distance travelled, we want to take the value of α which maximises $\tan\alpha$, so we can take $\alpha = 80.67°$.

Now from Part (ii):

$$\tan 80.67° \geq \frac{(80)(1)}{(80+r)r}$$
$$6.09(r+80)r \geq 80$$
$$r^2 + 80r - 13.14 \geq 0$$

Now solving for the minimum positive value of r using the quadratic formula:

$$r = \frac{-80 \pm \sqrt{80^2 - 4(13.14)}}{2}$$
$$\approx 0.16 \text{ m.}$$

Thus, the closest point is 0.16 m from the wall, or 80.16 m from the point O.

END OF EXAMINATION

Student Name: ________________________________

Teacher: CHEN / QUINNELL

Section I: Multiple Choice Answer Sheet

Completely fill in the most correct answer

1	A ⬭	B ⬭	C ⬭	D ⬭
2	A ⬭	B ⬭	C ⬭	D ⬭
3	A ⬭	B ⬭	C ⬭	D ⬭
4	A ⬭	B ⬭	C ⬭	D ⬭
5	A ⬭	B ⬭	C ⬭	D ⬭
6	A ⬭	B ⬭	C ⬭	D ⬭
7	A ⬭	B ⬭	C ⬭	D ⬭
8	A ⬭	B ⬭	C ⬭	D ⬭
9	A ⬭	B ⬭	C ⬭	D ⬭
10	A ⬭	B ⬭	C ⬭	D ⬭